

Sabrina Abram, PhD

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Summary

Computational cognitive neuroscientist studying adaptive behavior in humans. My research combines reinforcement learning models with behavioral and neuropsychological experiments to understand **how agency shapes human behavior and how behavior changes when that agency is shared with other systems**. This research has informed human-in-the-loop systems and extends to fundamental questions of agency and alignment in AI systems.

Education

University of California, Berkeley , Postdoctoral Fellow Departments of Psychology and Neuroscience	2022–present
Stanford University , Visiting Scholar Department of Mechanical Engineering	2019–2020
Simon Fraser University , PhD Department of Engineering Science	2015–2021
Cleveland State University , BSc Summa Cum Laude Department of Mechanical Engineering; Division I Athlete	2011–2015

Experience

10+ years of hypothesis-driven and data-driven science

Cognition and Action Lab, University of California, Berkeley

Advisor: Rich Ivry, Focus: Agency and reinforcement learning

- Identified distinct cognitive biases in decision making driven by agency in action selection and action execution
- Developed reinforcement learning models that found dissociable neural mechanisms of agency in decision making
- Managed teams across Berkeley, Yale, and Carnegie Mellon, and presented results at 4 international conferences

Biomechatronics Lab, Stanford University

Advisor: Steve Collins, Focus: Human-in-the-loop optimization

- Analyzed longitudinal datasets of human-exoskeleton interaction to reveal explore-exploit strategies in humans
- Identified computational mechanisms and behavioral markers of learning during human-in-the-loop optimization
- Published findings in *Current Biology*, with an invited dispatch article discussing the broader impact of the work

Locomotion Lab, Simon Fraser University

Advisor: Max Donelan, Focus: Energy optimization in human locomotion

- Designed and built a robotic device to determine whether humans optimize energy as they learn new movements
- Developed computational models of energy optimization to identify how humans learn in high-dimensional spaces
- Awarded Canada's top doctoral fellowship for research and leadership; presented at 10 international conferences

Publications

In Preparation

1. **Abram SJ**, Bolo Dave L, Tsay JS, Wang T, Collins AGE, McDougale SD, and Ivry RB. Agency in action selection and action execution produce distinct biases in decision making.
2. **Abram SJ**, Tsay JS, Wang T, Li JJ, Collins AGE, McDougale SD, and Ivry RB. Differential roles of the cerebellum and basal ganglia in agency-dependent reinforcement learning.
3. Sukumar S, **Abram SJ**, Chirino-Pérez A, and Ivry RB. Cerebellar degeneration impairs volitional category switching in semantic fluency.

4. Cisneros E, Bol K, Collins AGE, **Abram SJ**, Ivry RB, and Tsay JS. No evidence for the role of the cerebellum in value-based decision making.
5. **Abram SJ**, Ortega J, Wang T, Whitney D, and Ivry RB. Cerebellar mechanisms of dynamic emotion tracking.

Journal Articles, Book Chapters, and Refereed Conference Proceedings

6. **Abram SJ**, Chirino-Pérez A, Wang T, and Ivry RB (Under Review). Cerebellar contributions to human social cognition. In *Advances in Neurobiology*. Springer.
7. **Abram SJ**, Tsay JS, Wang T, Collins AGE, McDougle SD, and Ivry RB (2025). [Agency in action selection and action execution produce distinct biases in decision making](#). *Proceedings of the 6th RLDM Conference*.
8. Cisneros E, Bol K, Collins AGE, **Abram SJ**, Ivry RB, and Tsay JS (2025). [No evidence for the role of the cerebellum in value-based decision making](#). *Proceedings of the 6th RLDM Conference*.
9. Chen Y, **Abram SJ**, Ivry RB*, and Tsay JS* (2025). [Indirect feedback hinders explicit sensorimotor adaptation](#). *Proceedings of the Royal Society B: Biological Sciences*.
10. **Abram SJ***, Tsay JS*, Yosef H, Reisman DS, and Kim HE (2025). [The detrimental effect of stroke on motor adaptation](#). *Neurorehabilitation and Neural Repair*.
11. Wang T, Avraham G, Tsay JS, **Abram SJ**, and Ivry RB (2024). [Perturbation variability does not influence implicit sensorimotor adaptation](#). *PLOS Computational Biology*.
12. van Mastrigt NM, Tsay JS, Wang T, Avraham G, **Abram SJ**, van der Kooij K, Smeets JB, and Ivry RB (2023). [Implicit reward-based motor learning](#). *Experimental Brain Research*.
13. **Abram SJ**, Poggensee KL, Sánchez N, Simha SN, Finley JM, Collins SH, and Donelan JM (2022). [General variability leads to specific adaptation toward optimal movement policies](#). *Current Biology*. Featured in Current Biology Dispatch: [Motor control: In constant pursuit of optimality](#).
14. Simha SN, Wong JD, Selinger JC, **Abram SJ**, and Donelan JM (2021). [Increasing the gradient of energetic cost does not initiate adaptation in human walking](#). *Journal of Neurophysiology*.
15. **Abram SJ**, Selinger JC, and Donelan JM (2019). [Energy optimization is a major objective in the real-time control of step width in human walking](#). *Journal of Biomechanics*.
16. Kashiri N, Abate A, **Abram SJ**, et al. (2018). [An overview on principles for energy efficient robot locomotion](#). *Frontiers in Robotics and AI*.

Skills

Technical Skills: Programming (Matlab, Python, R); Data analysis (mixed-effects models, permutation testing, bootstrapping); Reinforcement learning (parameter estimation and recovery, model simulation and comparison)

Science Communication: 14 international conference presentations spanning cognitive neuroscience, psychology, artificial intelligence, machine learning, and robotics, including invited talks and award-winning presentations

Mentorship: Formally supervised over 10 students (2 Honors) at Berkeley and Simon Fraser University; mentored students in STEM and Women in Science programs at University of British Columbia and Canadian Science Expo

Service

Reviewer: Brain; IEEE Transactions on Neural Systems & Rehabilitation Engineering; Journal of Biomechanics; Journal of Experimental Biology; Neuroscience; Communications Psychology; Experimental Brain Research, NPJ

Committee Member: Society for the Neural Control of Movement; Parkinson's Disease Research Symposium

Captain: NCAA Division I Women's Soccer; achieved program records in scoring, shutouts, and team GPA

Scholarships and Awards

(Selected)

New Investigator Scholarship , Society for the Neural Control of Movement	2024
Michael Smith Foreign Study Supplement , NSERC Canada	2019–2020
Vanier Canada Graduate Scholarship (\$150,000), NSERC Canada	2016–2019
NCAA Division I Women's Soccer Scholarship , Cleveland State University	2011–2015